Lesson 7.4 — The Tangent Function

Big idea: tan θ = sin θ / cos θ. It tells you the slope of the unit-circle radius line. Where cosine is zero, tangent has a vertical asymptote.

Quick review

Definition: tan θ = sin θ / cos θ.
Period: π (NOT 2π) — tangent repeats every half-circle.
Asymptotes: at θ = π/2 + nπ (where cos θ = 0).
Range: all real numbers.
Special values: tan 0 = 0, tan(π/4) = 1, tan(π/3) = √3, tan(π/6) = √3/3.

Worked example

Worked example. Find tan(2π/3).

Q2: sin = √3/2, cos = –1/2. tan = (√3/2)/(–1/2) = –√3.

Warm-ups

1. Find tan(0).
2. Find tan(π/4).
3. Where is tan(θ) undefined for 0 ≤ θ < 2π?

Practice

1. Find tan(π/3).
2. Find tan(5π/6).
3. Find tan(7π/4).

Challenge

1. Solve tan θ = 1 for 0 ≤ θ < 2π.
2. Show that $\tan^2\theta + 1 = \sec^2\theta$ where sec θ = 1/cos θ. (Hint: divide the Pythagorean identity by $\cos^2\theta$.)

Answer key — Lesson 7.4

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 0.
2. 1.
3. $\theta = \pi/2$ and $3\pi/2$ (cos = 0 there).

Practice

1. $\sqrt{3}$.
2. Q2: sin positive, cos negative. $\tan(5\pi/6) = (1/2)/(-\sqrt{3}/2) = -1/\sqrt{3} = -\sqrt{3}/3$.
3. Q4: sin negative, cos positive. $\tan(7\pi/4) = (-\sqrt{2}/2)/(\sqrt{2}/2) = -1$.

Challenge

1. $\theta = \pi/4$ (Q1) and $5\pi/4$ (Q3).
2. $\sin^2\theta + \cos^2\theta = 1 \Rightarrow$ divide by $\cos^2\theta$: $\tan^2\theta + 1 = 1/\cos^2\theta = \sec^2\theta$. ✓